MA289 Project 3

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1 Introduction

In this project, the effects of ridge regression (or ridge regularization) are explored, particularly regarding how ridge regression is affected by unstandardized inputs. Ridge regression is similar to ordinary least-squares regression (OLS), but an additional penalty is added to the error function equal to some constant multiple (lambda) of the sum of the squares of each feature coefficient. This has the effect of shrinking the values of coefficients, allowing the model to disregard irrelevant features.

To understand the behavior of ridge regression regarding features of varying scales, two synthetic datasets and one real housing dataset are explored, using ridge regression to perform prediction tasks on both standardized and unstandardized versions of each dataset. This analysis leads to the conclusion that ridge regression performs best when paired with feature standardization to avoid influence from feature scales.

2 Methodology

By exploring the behavior of ridge regularization across two synthetic datasets, the situations in which the method works well and breaks down can be demonstrated, and the degree to which data standardization remedies these issues can be investigated. From there, the same patterns and behaviors can be seen using real-world data, but not to the extreme level at which the synthetic data amplifies these properties.

2.1 Crafting Synthetic Data

To craft a synthetic dataset, a function is built such that a given number of input features are each assigned a scale and a weight. Each feature consists of data points randomly selected between 0 and the defined scale parameter of that feature, then output values are generated by summing the value of each feature multiplied by that feature's importance parameter, then applying normally distributed error to the sum. Fig. 1 provides an example dataset

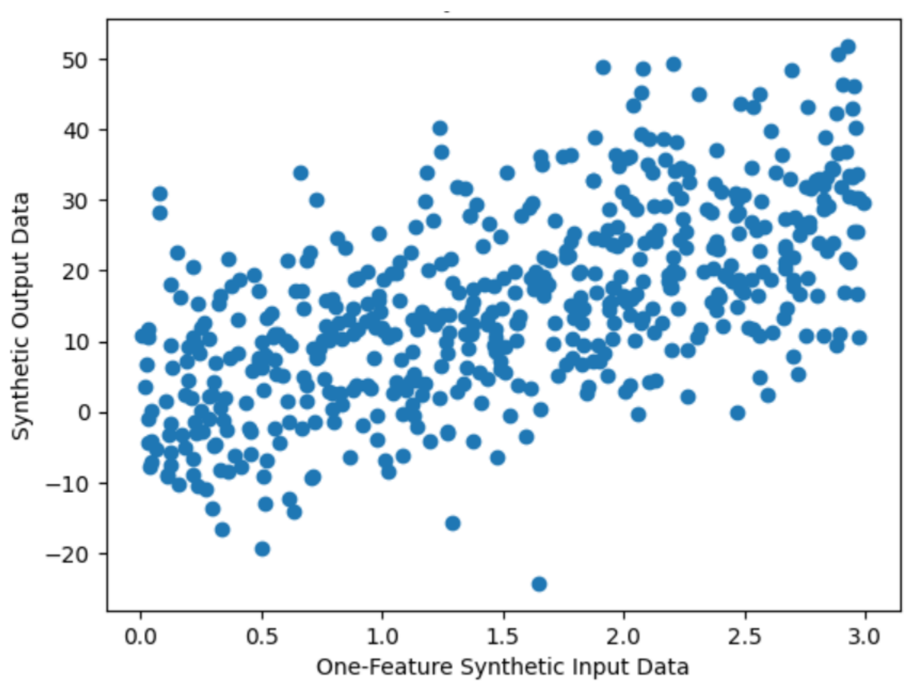

Figure 1: Demonstration of Synthetic Dataset Generation

generated by this method, comprised of one feature with a scale of 3 and an importance of 10.

The first synthetic dataset generated consists of 7 features, with the scale of one feature 10^7 times larger than the rest of the features, but the importance of that feature is 10^9 times smaller than the rest, meaning the feature is overall 100 times less important than the rest of the features when predicting the output. This dataset is designed to understand how ridge regression handles large, irrelevant features.

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2.2 Exploring Real Housing Data

A housing dataset is used to understand the behavior of ridge regression on real data, which consists of 16 input features (after categorical, incomplete, and totally irrelevant features are removed) to predict the sale price of a house. This dataset is selected because it provides a good mix of features at various scales, with small features including quality and number of cars that fit in the garage (single digit numbers) and large features including various square footage measurements (which consist of data points in the thousands). To understand these features, single linear regression is performed on each feature to understand its correlation with the sale price of a house, revealing a good mix of strong and weak predictors in the dataset.

3 Results

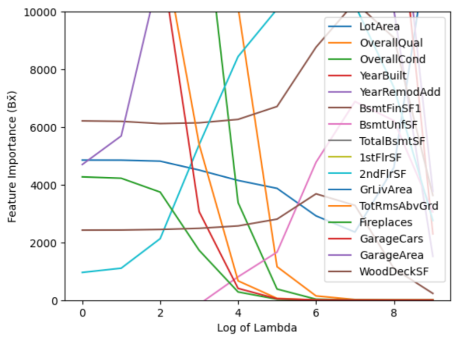
3.1 Ridge Performance on Synthetic Data

To investigate the difference in performance between OLS, ridge, and standardized ridge, prediction was performed on both datasets using all three models, with 80 percent of the data used for training and the other 20 percent used for model evaluation. When using the first dataset (with one large, irrelevant feature), all three models performed almost identically (mean squared errors of 27.64, 27.65, and 27.67), and all three correctly identified that the coefficient for the feature of interest should be significantly smaller than the rest of the coefficients. This demonstrates a strength of ridge regression in its baseline use case, as irrelevant parameters are effectively suppressed. However, when using the second dataset (with one small, important feature), ridge regression (with a lambda value of 0.01) significantly underperformed OLS; ridge yielded a mean squared error of 2293, compared to 1375 with OLS. This makes sense in the context of the significant feature's coefficient, which OLS sets 10^9 times larger than the rest of the coefficients (to offset the scale and importance differences), but ridge regression penalizes that large coefficient, setting it roughly equal to the rest of the coefficients (leave the small-scale feature to be pretty much ignored). This behavior is remedied when the data is first standardized, with standardized ridge performing similarly to OLS and demonstrating how ridge can only effectively prioritize small-scale features when data is standardized.

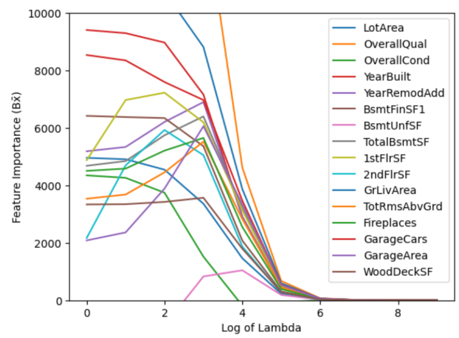
3.2 Ridge Performance on Real Housing Data

To understand how this issue of managing features of varying scales manifests in a real dataset, both unstandardized and standardized versions of the housing dataset were used to train 10 ridge regression models to predict the sale price of a house, with the value of lambda increasing exponentially for each successive model. Using the same 80/20 train-test split, both sets of models performed fairly well until lambda = 100 (both maintaining mean squared errors around $1.4 * 10^9$), but further increases to lambda led to precipitous decreases in the performance of both sets of models. While slightly better performance was maintained using the standardized models, what is important to note is the manner in which ridge regression is highly variable regarding which features are prioritized across lambda values, while standardized ridge regression suppresses features in a much more smooth and predictable manner as lambda increases.

This effect is demonstrated in figs. 2a and 2b, which highlight how the combinations of features which the model prioritizes is sensitive to the amount of regularization applied (lambda) without standardization but not sensitive to lambda with standardization.



(a) Without Standardization



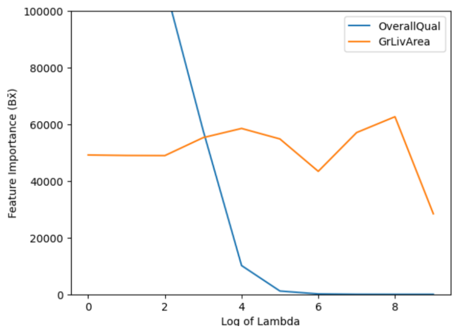
(b) With Standardization

Figure 2: Ridge Regression Selection of Features Across Lambda Values

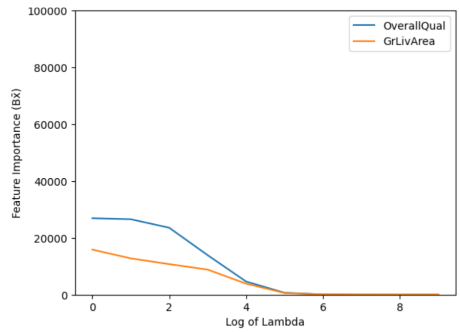
To understand why this is the case, it is helpful to look at specific feature behaviors. One such behavior is the interaction between features defining the quality of the house (a number between 1 and 10), and the amount of living area (between 0 and 5000 square feet). Both features are pretty good indicators of sale price, with the quality being slightly better predictor; a single linear regression between quality and price returns an r-squared of 0.63, compared to 0.50 for living area. As fig. 3a demonstrates, quality is initially factored into the model more heavily than living area, but with more regularization, living area (the worse predictor) is favored. This provides a good real-world example of the findings from the synthetic dataset; ridge regularization struggles to prioritize small-scale but important features, as the cost function punishes the high coefficients associated with these features. Fig. 3b shows how this issue is resolved with standardization, as both features are suppressed at similar rates. Figs. 4a and 4b show a similar phenomenon with two more closely linked features, as the number of cars that fit in the garage (a slightly better predictor) is deprioritized in deference to the garage area (a feature with a larger scale) as the amount of regularization increases, but this flaw is not present when data is first standardized.

4 Conclusion

Overall, this analysis of ridge regularization with and without standardization as compared to OLS on both synthetic and real datasets has shown that while ridge can be an effective tool for suppressing irrelevant features, it is best paired with data standardization to avoid unintentional suppression of features that are important but of a smaller scale. By understanding these strengths and

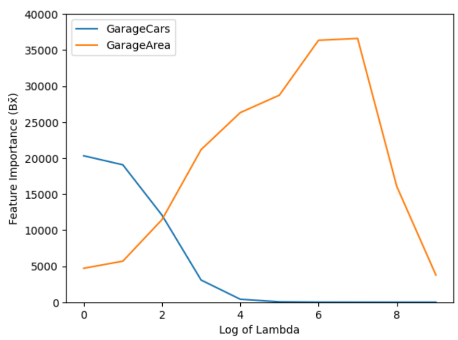


(a) Without Standardization

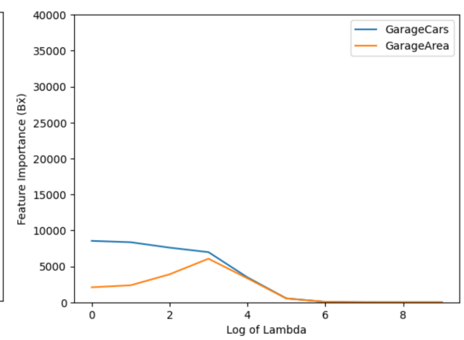


(b) With Standardization

Figure 3: Ridge Regression Prioritization of Quality and Living Area Across Lambda Values



(a) Without Standardization



(b) With Standardization

Figure 4: Ridge Regression Prioritization of Garage Cars and Garage Area Across Lambda Values

weaknesses of the method, it can be better utilized to build stronger predictive models using linear regression.

5 Bibliography

[1] “House Prices - Advanced Regression Techniques.” Accessed: April 15, 2026. [Online]. Available: https://kaggle.com/house-prices-advanced-regression-techniques